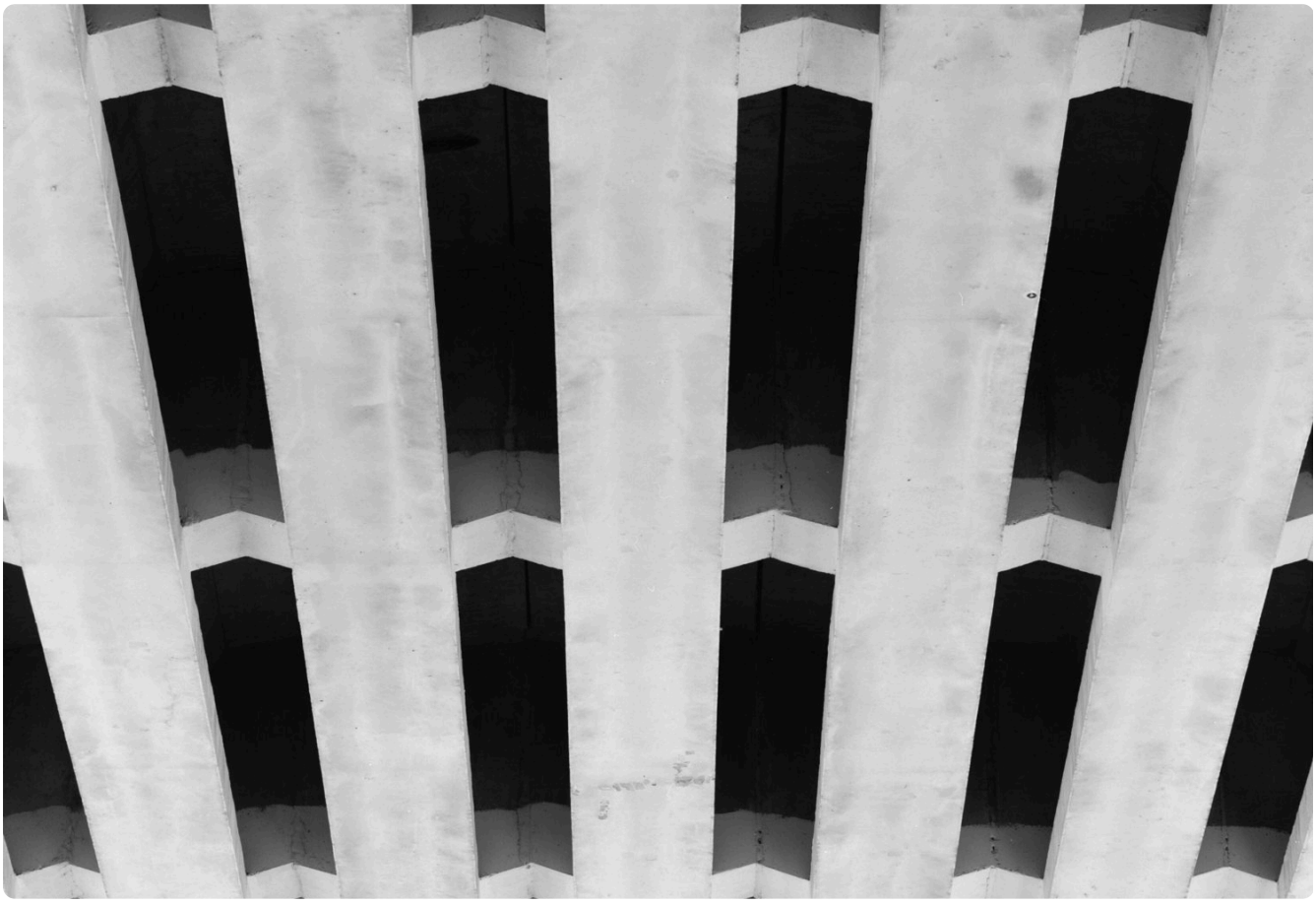
DESIGN COMPUTED · FREE GUIDE

Squeeze it before it's loaded: how post-tensioning lets concrete span

The free guide to the reel on post-tensioned concrete and America's first prestressed bridge: why concrete needs steel that pulls, how tendons are stretched and locked, and a book-squeezing experiment you can do at your desk.



The problem: concrete hates being pulled

Concrete is strong when it is squeezed but cracks easily when it is pulled. A beam on two supports bends under load: the top is squeezed and the bottom is pulled. So the bottom is where plain concrete cracks first.

Ordinary reinforced concrete puts steel bars in the bottom to hold the cracks together. That works, but the concrete still cracks and the beam still has to be deep and heavy to span far.

The idea: squeeze it first

Prestressed concrete is compressed during production so that it already resists the pulling forces it will meet later. The tendons are usually high-strength steel, though carbon or aramid fibre can also be used.

In post-tensioning, tendons run inside ducts cast in the concrete. After the concrete has cured, the tendons are stretched with jacks, pulled through anchorages that press on the concrete, and locked off. The locked tendon leaves a lasting compressive force in the beam.

Curved tendons carry load too

The tendon profile can be curved: low in the middle of the span, high at the ends. A curved stretched cable pushes the beam up from inside, so it counters the load that will press down later. In bonded post-tensioning, the ducts are then filled with grout to protect the steel from corrosion.

Built examples

Walnut Lane Memorial Bridge, Philadelphia (1950)	The first major prestressed concrete beam bridge built in the United States, designed by Belgian engineer Gustave Magnel: three spans, each of 13 girders 160 feet (49 m) long, with four wire cables post-tensioned in each.
Eugene Freyssinet's patent (1928)	Prestressed concrete was patented by Eugene Freyssinet in 1928.
Typical tendons	Building tendons usually have 2 to 6 strands each; large dams use tendons of up to 91 strands.

Try it yourself: lift books by squeezing

1. Take 8 to 10 paperback books and line them up on a table like a row of dominoes.
2. Try to lift the whole row by holding only the two end books lightly. The row sags and falls apart: that is plain concrete.
3. Now press the two ends toward each other firmly and lift again. The row stays rigid and you can carry it, because the squeeze keeps every joint closed.
4. Hang a small bag from the middle book while still squeezing. See how much more load the squeezed row takes.
5. Release the squeeze a little and watch where it opens first: at the bottom. That is where a real beam cracks.

The rule to remember

Put the stress in before the load arrives, and the load has less to break.

Default answer	Add more material until it doesn't crack.
Form-finder answer	Add a force that works against the failure.
Check	Ask: what could I pre-load in this design so that the real load starts from a better place?

Go further

Prestressed concrete (Wikipedia)
https://en.wikipedia.org/wiki/Prestressed_concrete

Walnut Lane Memorial Bridge (Wikipedia)
https://en.wikipedia.org/wiki/Walnut_Lane_Memorial_Bridge

Federal Highway Administration: bridges
https://highways.dot.gov/bridges-and-structures

Credits
Cover photo: A. Pierce Bounds, HAER, Library of Congress, public domain.
Facts: Wikipedia (Prestressed concrete, Walnut Lane Memorial Bridge).

Made by Design Computed · Follow for more · @designcomputed